BMC Health Services Research

RESEARCH Open Access

Towards mitigating overcrowding in urban Indian healthcare facilities: stakeholder perception analysis for existing and potential new patient referral mechanisms

Najiya Fatma[1*], Kaveri Kala[2] and Varun Ramamohan[1]

Abstract

Background Public healthcare delivery in India faces several operational challenges, including congestion and long wait times at higher-level facilities and low utilization of lower-level facilities. Effective referral mechanisms can help address these issues. However, before designing and implementing potential new referral mechanisms, it is essential to understand patient and provider views. The objective of this study is to quantitatively assess patient and provider perceptions of both existing and potential new referral mechanisms and their association with patient socioeconomic and demographic attributes.

Methods A cross-sectional survey was conducted to record and examine patient referral mechanisms – from patient and provider perspectives - currently operational at public healthcare facilities in an urban Indian district. Patient and doctor perceptions regarding potential new referral mechanisms were also assessed, such as medical referral with noncompliance penalties for out-of-turn visits to higher-level facilities, and operational referral, which involves referring patients to same-tier facilities with lower patient loads. Multinomial logistic regression was used to identify statistically significant associations between patient perceptions and socioeconomic and demographic attributes.

Results Survey results provided quantitative evidence of nonadherence to existing referral mechanisms by a significant proportion of patients, and logistic regression analyses showed statistical associations between patient socioeconomic and demographic variables and their willingness towards implementation of potential new referral mechanisms.

Conclusions Based on study findings, existing referral mechanisms can be strengthened, and potential new referral mechanisms, with appropriate refinements, can be introduced to mitigate overcrowding, care provision delays, and other operational challenges in Indian healthcare delivery.

Keywords Healthcare facility utilization, Overcrowding, Patient referral, Operational referral, Stakeholder perception, Good health and wellbeing

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Background

The unstructured manner in which referrals currently operate in the Indian public healthcare delivery system has led to overcrowding of tertiary hospitals with patients whose conditions could be managed at primary or secondary facilities, which often remain under-utilized [1, 2]. This results in operational inefficiencies such as long wait times prior to receipt of care, poor patient outcomes, and strain on the overall hierarchical healthcare system. Similar issues have been reported in other developing countries in Southeast Asia and Africa [3, 4]. The overall hierarchy of referral pathways in the Indian public healthcare system is summarized in Fig. 1.

A key contributor to overcrowding and delays in care provision at urban tertiary care public facilities and underutilization of lower-level public facilities is the widespread bypassing of primary and secondary care facilities, with patients seeking direct access to specialists at higher-level facilities [5–7]. Factors driving such bypassing include the perception of getting better quality of care at tertiary facilities, lack of clearly defined care pathways and referral mechanisms, and absence of referrals to lower-level facilities for patients with minor illnesses visiting higher-level facilities, etc. Bypassing also strains the healthcare system and reduces effective communication between primary care providers and specialists, leading to poorer mortality and morbidity-related outcomes in the long run [8]. In urban India, nearly 70% of healthcare visits occur in the private sector, where the proliferation of small clinics often run by providers with limited formal training, further complicates coordinate across different levels of the public system [9].

Integration of effective referral mechanisms - vertical (medical referral), horizontal (operational referral), and telemedicine-based referral - are essential to direct patients appropriately within the healthcare network [10–13]. Medical referral (MR) involves directing a patient from a lower-level facility (e.g., primary care facility) to a higher-level facility (e.g., tertiary care facility) due to insufficient expertise and/or equipment required to manage the patient's condition [11]. Operational referral (OpR) diverts patients to a facility of a similar level due to the unavailability of resources (e.g., doctors, beds) at the facility of first visit, typically due to overcrowding or congestion [12]. Telemedicine based referral with the use of information technology (IT) can support both MR and OpR across geographies [14]. Successful implementation of these mechanisms require electronic medical records (EMR) systems to maintain complete patient information across the healthcare delivery network. These referral mechanisms also have implications for programs like the Pradhan Mantri Jan Arogya Yojana (PM-JAY), a national health insurance scheme launched in 2018 to provide financial protection and improve healthcare access. The integration of electronic medical records with PM-JAY can improve the tracking of patients' claims and treatment history, reducing fraud and streamlining the reimbursement process for empanelled hospitals [15].

Evidence from other developed countries highlighted the benefits of implementing these effective referral mechanisms: EMR systems improved timely medical care, imposition of noncompliance penalties reduced bypassing, OpR minimized average time to receive care, and telemedicine-based consultation services (telestroke, teleophthalmology tele-general health) improved patients' health, especially in remote regions [16, 17]. In India, however, formal mechanisms for implementation of these referral mechanisms - such as OpR, noncompliance penalties within MR, and others - are lacking within the public healthcare network. Further, there is limited evidence regarding stakeholder perceptions of how effective these referral mechanisms might be if they were implemented within the network.

Globally, stakeholder involvement is recognized as essential for effectively designing health innovation planning processes. Stotter and colleagues surveyed stakeholders regarding a new scope of practice in primary healthcare in New Zealand, while online surveys

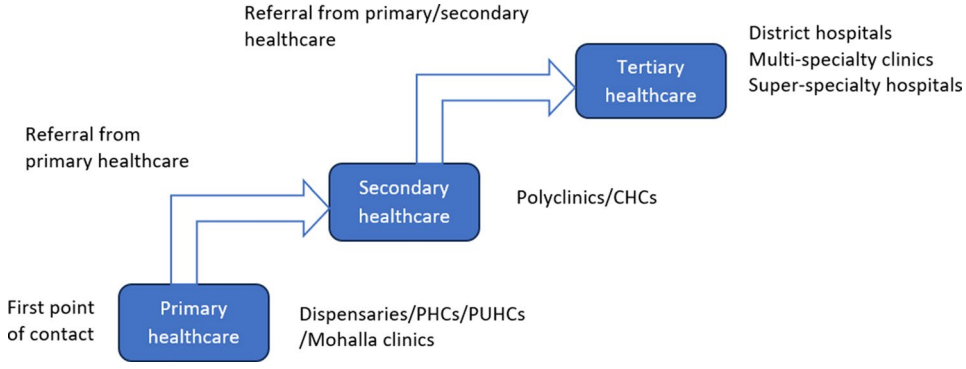


Fig. 1 Hierarchical levels of the Indian public healthcare referral system (PHCs: Primary health centres; PUHCs: Primary urban health centres; CHCs: Community health centers)

of telemedicine providers were conducted in Egypt to estimate perceived savings and reductions in healthcare delivery wait times, and to identify implementation facilitators and barriers [18, 19]. Franco-Trigo and colleagues summarized stakeholder analysis approaches and reporting guidelines in health innovation planning processes worldwide [20].

Designing referral mechanisms such as those discussed above requires knowledge regarding stakeholder perceptions of existing mechanisms. Further, their implementation requires an appropriate (dis)incentive structure for patients, which may, for example, take the form of non-compliance penalties. However, documentary evidence on referral mechanisms currently implemented across hierarchical healthcare delivery networks in developing countries and emerging economies, including India, is limited or non-existent. It is also crucial to understand the perceptions of key stakeholders associated with the healthcare delivery system so that effective referral systems that meet the diverse needs and expectations of key stakeholders are instituted. Therefore, in this study, we record and examine the perceptions of patients and doctors regarding existing as well as potential new referral mechanisms.

We conducted a cross-sectional survey among patients and doctors at public primary and secondary healthcare facilities in the South-west Delhi district in India to record and analyse: (a) their experiences with existing referral mechanisms, and (b) their perceptions and willingness towards the implementation of potential new referral mechanisms, which include MR with non-compliance penalties (via research question *R1* and *R2*), OpR (via *R3*), centralised electronic medical records (EMR) system (via *R4*), and telemedicine-based consultation and referral (TMR) (via *R5*). The research contributions of our study lie in addressing the above gaps via the generation of quantitative evidence associated with stakeholder perceptions of referral mechanisms. We also analysed demographic and economic factors significantly influencing patient perceptions using multinomial logistic regression (MLR). Key concerns of doctors regarding the feasibility of implementation of the new referral mechanisms, which may need to be addressed to ensure effective deployment of these mechanisms, were also recorded.

Methods

Study design and survey questionnaire design

A cross-sectional survey was conducted at twenty-two primary and secondary public healthcare facilities from December 2019 to April 2022 in the South-west Delhi district in India. South-west Delhi has a well-developed healthcare system with both public and private facilities to provide accessible and quality healthcare. We chose South-west Delhi as the study area owing to two reasons: (a) availability of significantly larger number of primary and secondary public healthcare facilities in comparison to other districts, and (b) proximity to the authors' institute. We had to suspend the survey twice due to the COVID-19 pandemic and gathered participant responses within a total duration of six months. The Directorate of Health Services, Government of National Capital Territory (NCT) of Delhi, provides healthcare services via a network of dispensaries, primary urban health centres (PUHCs), polyclinics, and multi- and super-specialty hospitals. Dispensaries and PUHCs are the first points of medical contact to provide treatments for ailments of mild to moderate severity. Patients requiring specialized care are referred to a linked higher-level facility by general physicians at dispensaries and PUHCs. Multi-specialty polyclinics provide secondary healthcare via outpatient (OPD) consultations by specialists from the internal medicine, gynaecology, paediatrics, orthopaedics, and ophthalmology disciplines, among others. Multi- and super-specialty hospitals provide tertiary care in all medical specialties. Twenty-two facilities were randomly selected (17 dispensaries out of 23, 2 PUHCs out of 7, and 3 polyclinics out of 4) for administering surveys – separate questionnaires for patients and doctors - among the participants.

The survey was administered by the first author (female) through in-person visits to the public healthcare facilities in South-west Delhi. The purpose of administering the survey was explained to all participants, who provided written informed consent, and participation was voluntary. Responses were recorded on paper-based questionnaires available in Hindi and English. A pilot survey was conducted first to refine the questionnaire based on observed errors, participant responses, and the total survey completion time. All responses were collected using a consistent procedure across facilities, de-identified, securely stored in Excel, and carefully maintained to ensure data quality. No personally identifiable information was collected from patients. All questionnaires with incomplete responses were excluded during data cleaning, resulting in no missing data in the final dataset and allowing for a complete-case analysis approach.

The patient survey questionnaire had three sections to record: (a) their experiences within the referral system currently operational at public healthcare facilities; (b) their perceptions regarding new referral mechanisms that could potentially be deployed in the healthcare system; and (c) the demographic and socioeconomic attributes of patients. The doctors' survey questionnaire had two sections to record: (a) operational information regarding healthcare delivery and existing referral mechanisms at their facility, and (b) their perceptions regarding new referral mechanisms. The second section included

questions to record concerns of doctors regarding the feasibility of deployment of the new mechanisms. Participants rated their perception of potential new referral mechanisms on a nominal Likert scale with three independent categories: (a) favourable (willingness to accept), (b) favourable with reservations (willingness to accept with few concerns – participant concerns were separately noted), and (c) unfavourable (unwillingness to accept). A 3-category likert is preferable as it reduces participants' cognitive load compared to a 5-category scale and facilitates easier interpretation in inferential analyses. The survey questionnaires can be viewed here: < shorturl.at/mIOY2>. The questions were primarily close-ended and straightforward, minimizing the risk of discomfort or intimidation. To reduce social desirability bias, the survey was administered in a neutral manner without judgmental framing, and no questions directly criticized the performance of specific facilities or providers.

Data collection

Based on responses collected from the pilot survey and with an error margin of 5% and assuming a 95% confidence level, four hundred and forty nine (449) patients were recruited after considering a 90% response rate, 95% eligibility rate effect, and a design effect of 1. However, responses from all the doctors present in the surveyed facilities during our visits were recorded due to their relatively smaller number. Categorical responses were summarised using frequencies and continuous responses were summarised using means and standard deviations.

Quantitative analysis via multinomial logistic regression

Patient willingness for the implementation of referral mechanisms was modelled using MLR to predict the probabilities of choosing *favourable* and *favourable with reservations* categories of the response variables, given a set of their demographic and socioeconomic attributes (Fig. 1) to the baseline *unfavourable* category. The reference category for the gender variable was 'male'; 'between 18–29' years for age; 'married' for marital status; 'no formal education' for the education level variable; 'up to USD 11,500' for the annual income level; 'student' for the employment status variable; and '0' for number of children. Associations between variables were quantified using odds ratios (OR) and variables with p-values < 0.05 were considered as significantly influencing the choices of patients. All statistical analyses were conducted via the R 4.1.2 software [21].

Before conducting MLR analyses, the research question and the corresponding research hypothesis for each referral mechanism were formulated. The formulation for the MR mechanism is presented below as an example.

Research question 1 (R1) What socioeconomic and demographic factors are associated with the perception of patients for the implementation of medical referral mechanisms at public healthcare facilities?

The null and alternate hypotheses associated with *R1* are as follows:

$\boldsymbol{H_o}$ Socioeconomic and demographic variables do not influence patient perceptions of the implementation of the medical referral mechanism at public healthcare facilities.

$$H_o : \beta_1 = \beta_2 = ... = \beta_k = 0$$

$\boldsymbol{H_a}$ Socioeconomic and demographic variables influence patient perceptions of the implementation of the medical referral mechanism at public healthcare facilities.

$$H_a : \beta_1 \neq \beta_2 \neq ....\neq \beta_k \neq 0$$

Similar to *R1*, we formulated *R2* (implementation of noncompliance penalties), *R3* (OpR mechanism), *R4* (EMR system implementation), and *R5* (TMR mechanism). These are listed below.

R2 What socioeconomic and demographic factors are associated with the perception of patients regarding the implementation of noncompliance penalties within the medical referral mechanisms at public healthcare facilities?

R3 What socioeconomic and demographic factors are associated with the perception of patients regarding the implementation of operational referral mechanisms at public healthcare facilities?

R4 What socioeconomic and demographic factors are associated with the perception of patients regarding the EMR system implementation at public healthcare facilities?

R5 What socioeconomic and demographic factors are associated with the perception of patients regarding the implementation of telemedicine-based consultation and referral at public healthcare facilities?

The data was also checked for the presence of multicollinearity. Higher correlation coefficients were observed for the age, employment status, and number of children variables with the marital status variable; however, removing the marital status variable had limited impact on model parameter estimates and Akaike information criteria (AIC) scores. Hence the variable was retained in the analysis.

Results

We begin by summarizing key operational information of the surveyed public healthcare facilities. Following this, demographic information about the surveyed patients and their experiences with existing referral mechanisms are described. Next, we present multinomial logistic regression results that assess statistical associations between patient attributes and their perceptions regarding the implementation of different potential new referral mechanisms. The feedback provided by doctors regarding the feasibility of potential new referral mechanisms within the existing health facility network is summarised at the end of this section.

Healthcare facility operational information

From each surveyed public healthcare facility, operational information was collected on average daily patient load, the average number of referrals made by healthcare providers, reasons for referral, whether referral records were maintained and the average distance to referral facility to understand the patient load and the existing referral mechanisms. Based on data collected from personnels (doctors, nurses, and other healthcare staffs) at the surveyed facilities and patient record registers, significant variations were observed in the average patient load across the healthcare facility network. For example, the average patient load at polyclinics was higher (210 patients/day) in comparison to that of dispensaries (190 patients/day) and PUHCs (130 patients/day), while the minimum and maximum daily average outpatient loads across dispensaries varied between 60 and 200.

Regarding existing referral operations, on average, 3–5 patients were referred every day, with childbirth delivery being the most common reason for referring patients to higher-level facilities. This was mainly due to the unavailability of labour beds at the surveyed facilities. Approximately three-fourths of the surveyed doctors maintained records of referral attendance. The distances between the medical facility of the first visit and the referral destination facility ranged between 10 and 12 km. More details, including reasons for referral, are provided in Table 1.

Patient's survey: descriptive analyses

Demographic profile of the surveyed patients

A brief profile of the sociodemographic and economic characteristics of the surveyed patients is presented in Table 2. Nearly half of the respondents (~ 50%) belonged to lower- and middle-income groups (annual income

Table 1 Summary of referral mechanisms currently followed in the surveyed facilities

Facility Number	# patients seen per day	# patients referred per day	Referral reason	For what conditions patients are referred?	Referral record	Distance between referral and the source facilities
1 (Dispensary)	100–120	6–7	SS	Spine issue; Injury; CC	RRNM	10 km
2 (Dispensary)	50–60	1–2	SS, OT	Hypertension; Seasonal flu; CC	RRM	DDU (8–10 km) PH also (if patients ask) (4–5 km)
3 (Dispensary)	150–180	2–3	SS	Dog bite: surgery; CC	RRM	10–12 km
4 (Dispensary)	80–90	5–6	SS	Breathlessness; stomach-ache	RRM	6–7 km
5 (Dispensary)	130–150	8–10	SS	CC	RRM	10–12 km
6 (Dispensary)	60–70	0	NA	CC	RRNM	NA
7 (Dispensary)	180–200	10–15	SS, OT	Severe back pain; inaction	RRM	12–13 km
8 (Dispensary)	100–120	4–5	SS	Mental disorder; diabetes; CC	RRM	7–8 km
9 (Polyclinic)	200–210	10–15	SS	Chest pain; heart ailments; flu	RRM	10–12 km
10 (Dispensary)	180–190	3–5	SS	Severe injury; CC	RRM	5–6 km
11 (Dispensary)	100–150	4–6	SS	Swelling of fingers	RRM	9–10 km
12 (Dispensary)	80–100	3–7	SS, OT	Reproductive issue; appetite loss; CC	RRNM	9–10 km
13 (Dispensary)	90–100	5–6	SS	Dental cavity; farsightedness; CC	RRM	8–10 km
14 (Dispensary)	70–100	3–5	SS	Breathlessness; CC	RRM	6–7 km
15 (Dispensary)	110–120	6–8	SS	CC; specialized treatments	RRM	8 km
16 (PUHC)	120–130	7–8	SS	CC; specialized treatments	RRM	12–15 km
17 (Dispensary)	170–180	Around 10	SS, OT	Surgeries; requirement of super specialized treatments	RRNM	7 km
18 (Dispensary)	120–160	5–6	SS	Measles; fever; CC	RRNM	10 km
19 (Polyclinic)	140–160	7–8	SS	Acute and severe fracture; surgery	RRM	10 km
20 (PUHC)	80–100	3–4	SS	Migraine; CC	RRM	DDU – 8 km PH (if patients insist) – 5 km
21 (Polyclinic)	160–190	8–9	SS	CC; high fever	RRM	About 10 km
22 (Dispensary)	100–110	5–6	SS	Spine problem; injury	RRNM	10 km

SS: specialist more suited to handle patient's condition; OT (e.g., patient's insistence for' referral): Other reasons; RRM: referral register is maintained; RRNM: referral register is not maintained; CC: childbirth case; NA: not applicable. DDU: Deen Dayal Upadhyaya hospital (public hospital); PH: private healthcare facility

Table 2 Participant profile characteristics at South-west Delhi public healthcare facilities

Patient profile*		Percentage (out of 100)
Gender	Male (x_{11})	29.33 (132)
	Female (x_{12})	70.67 (317)
Age	Between 18–29 (x_{21})	36.67 (164)
	Between 30–39 (x_{22})	29.55 (132)
	Between 40–49 (x_{23})	14.67 (66)
	Between 50–59 (x_{24})	11.78 (54)
	60+ (x_{25})	7.33 (33)
Marital status	Married (x_{31})	77.56 (348)
	Unmarried (x_{32})	22.44 (101)
Highest level of education	No formal education (x_{41})	18.88 (85)
	Upto 10th grade (x_{42})	16.22 (73)
	Upto 12th grade (x_{43})	17.77 (80)
	Undergraduate degree (x_{44})	33.11 (148)
	Post-graduate degree (x_{45})	14.02 (63)
Annual Income	Prefer not to say (x_{52})	24.22 (109)
	Upto USD 11,500 (x_{51})	26.67 (120)
	USD 11,500 – USD 23,000 (x_{53})	22.89 (103)
	USD 23,000 – USD 46,000 (x_{54})	14.67 (66)
	Above USD 46,000 (x_{55})	11.55 (51)
Occupational status	Student (x_{61})	15.56 (70)
	Homemaker (x_{62})	38.44 (172)
	Employed (x_{63})	36.67 (165)
	Unemployed (x_{64})	9.33 (42)
Number of children	0 (x_{71})	30.44 (137)
	1 (x_{72})	22.01 (99)
	2 (x_{73})	29.11 (131)
	3+ (x_{74})	18.44 (82)

*Generated from a field survey of 449 respondents. Education up to 10th grade and 12th grade are significant milestones in the Indian school education system and are formally recognized by all education boards. We converted annual income in INR to US dollars after adjusting for purchasing power parity. With regard to occupational status, students consisted of all those who were pursuing educational qualifications corresponding to the levels within the "Highest level of education" variable on a full-time basis. NA: not available. The notations in parentheses represent the abbreviated variable names used in the multinomial logistic regression

between 0 and 23,000 USD), consistent with previous findings that public healthcare facilities are predominantly utilized by lower-income populations. These results reaffirm that poorer groups rely on public sector hospitals and health services to a greater extent than wealthier groups [22].

Referral mechanisms at public healthcare facilities

Approximately 36% of the surveyed participants consulted other healthcare providers after a first visit to a healthcare facility nearby to their location/residence. Patients were referred to other facilities either by their healthcare providers or chose self-directed referral. Self-directed referral involves patients referring themselves to higher-level facilities without the explicit

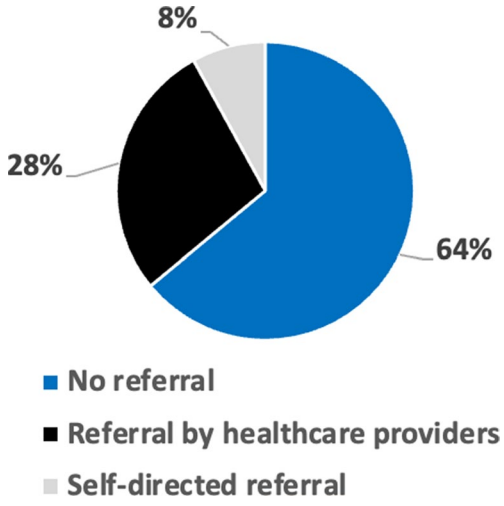


Fig. 2 Referral of patients to other facilities

recommendation of care providers. Out of 449 participants, 28% were referred previously to other facilities by healthcare providers (Fig. 2).

Among those referred by healthcare providers, nearly 98% were referred to higher-level facilities due to the requirement of specialized care. However, approximately 53% of the self-directed referral patients sought care from other facilities primarily due to congestion and overcrowding. Long wait time, lack of medical resources, and occasionally, unprofessional behavior and poor bedside manner of healthcare providers aggravated self-directed referral. A significant proportion of patients who chose self-directed referral were graduates and postgraduates (nearly 76%), belonging to the higher income category (44%), and were employed (67%). Healthcare providers from public facilities primarily referred patients to higher-level public facilities (nearly 86%). A few practitioners also referred patients to private facilities, mainly due to unavailability of medical resources at the higher-level public facilities and occasionally due to insistence from patients.

A few patients were only verbally referred to higher-level facilities by the doctors - i.e., without a referral slip or indication on their prescription. Among patients opting for self-directed referrals, a majority (82%) visited private healthcare facilities regardless of the substantially higher overall treatment costs due to the perception of increased convenience, ease of admission, and smaller delays before receiving treatment. Travel times to public referral healthcare facilities (average = 54.16, S.D. = 31.97) appeared to in general be higher when compared to those for private facilities (average = 30.21, S.D. = 11.39) (see Fig. 3).

The wait time before consulting the doctor at the destination public healthcare facility was 65.04 min (minimum = 30, maximum = 180, S.D. = 41.89), whereas wait

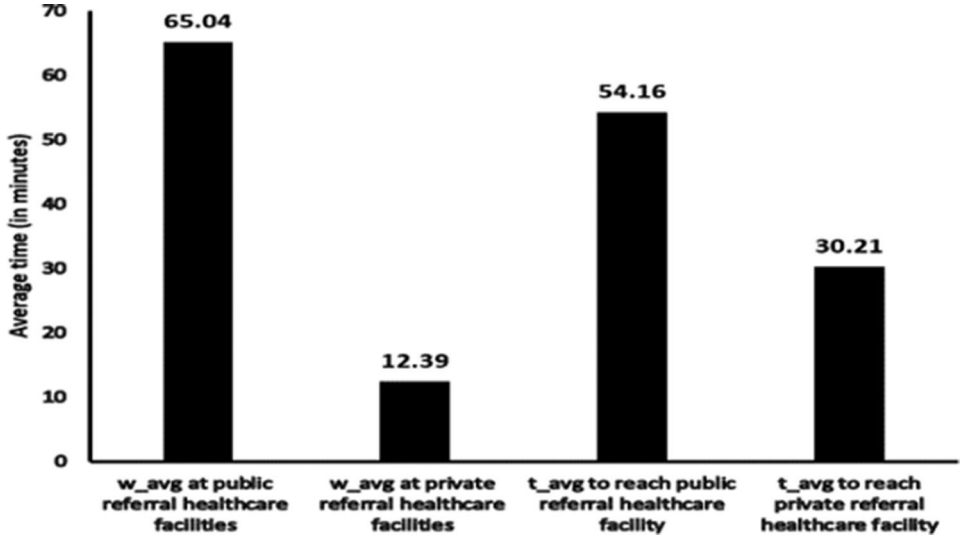


Fig. 3 Average wait time and average travel time at/to referral healthcare facilities. Note: w_avg = average wait time; t_avg = average travel time

time at primary-level healthcare facilities such as dispensaries and PUHCs (i.e., facilities where a patient is likely to visit first) varies between 1 and 2 min before receiving outpatient care. This is indicative of significantly higher patient loads at specialized public healthcare facilities, which could partly be attributed to bypassing of lower-level facilities. A few patients were also denied care at the public referral facility owing to high patient loads and had to visit another health facility to receive care. Substantially smaller wait time at private healthcare facilities were reported by the surveyed patients, as shown in Fig. 3.

Further, at the referral facilities, approximately 47% patients were re-examined regardless of the availability of a recent investigation report, with private healthcare providers primarily asking for test repetitions (76% of time out of 100%). This has both operational and economic implications, increasing the utilization of diagnostics providers, and also likely leading to an increase in follow-up consultations.

Feedback on potential new referral mechanisms

Based on the observations reported in Sect. 3.2.2, OpR mechanisms were not formally deployed at the surveyed healthcare facilities. The surveyed facilities also did not have the technological infrastructure for storing the medical histories of the patients and did not offer telemedicine-based consultation or referral services. This finding underscored the need for surveying patients on their willingness towards implementation of potential new referral mechanisms at public healthcare facilities.

Patients' views were analysed regarding multiple aspects of the implementation of new referral mechanisms at public healthcare facilities. First, responses were collected regarding their perceptions of a reasonable wait time before consulting doctors at a public healthcare facility. Per the surveyed patients, the average reasonable wait time was 27.09 min (minimum = 2; maximum = 120; S.D. = 17.07). Patients referred - either by healthcare providers or who chose self-directed referral - to private healthcare facilities reported a lower threshold for reasonable wait time in comparison to patients referred to public facilities. Patient perceptions were also recorded regarding reasonable travel times between the referral and the referring facilities. The perceived reasonable travel time to public healthcare facilities (average = 39.12 min, S.D. = 18.92) was marginally higher than that for private facilities (average = 34.45 min, S.D. = 19.64).

Patients' views on the implementation of new referral mechanisms at public healthcare facilities were recorded, and this is presented in Fig. 4. Approximately 65% of the participants favoured the implementation of MR mechanisms that ensures higher-level facilities are visited only via referral from the lower-level care providers. Others (21%) desired freedom in choosing healthcare facilities or care providers. Regarding the implementation of a centralised EMR system, a relatively small proportion of patients (14%) – mostly female – had concerns about the confidentiality of health-related information on the centralised EMR system. However, a majority (76%) of all patients favoured its implementation given that it might help in better explanation of medical conditions and lower likelihood of losing test reports.

Important concerns among participants regarding the implementation of new referral mechanisms were also recorded. High and consistent quality of healthcare across all facilities in the referral network was ranked as the most important concern followed by reasonable wait time and reasonable travel time between healthcare

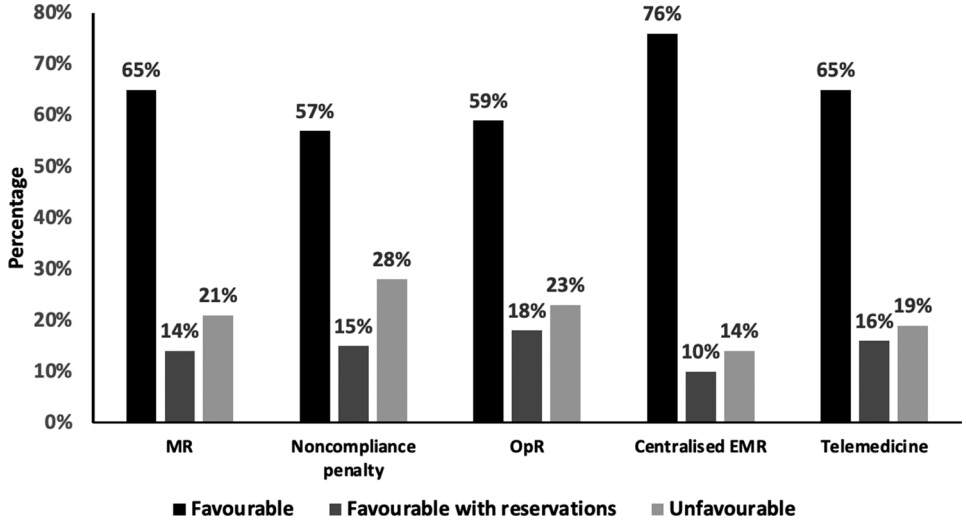


Fig. 4 Patients' feedback on potential new referral mechanisms

facilities. Other key concerns raised by patients were cleanliness at healthcare facilities, reasonably higher consultation durations than those encountered in current practice, and professional and cordial behavior of healthcare providers including doctors, staff nurses, lab technicians, and pharmacists.

Patient survey: Inferential analyses

To answer *R1*, the 'favourable' and 'favourable with reservation' categories of the response variable were regressed with the 'unfavourable' category as the reference. Patient attributes reported in Table 1 were treated as independent variables with discrete categories. Out of seven independent variables, four variables namely marital status, age, employment status, and annual income level were found to have statistically significant levels of association with the perceptions of patients regarding the implementation of the medical referral mechanism (Eq. 1).

$$\begin{aligned} &log\left(\frac{P(Y=favourable)}{P(Y=Unfavourable)}\right)_{R1} \\ &= 3.11 - 1.49x_{22} - 1.63x_{23} - 1.36x_{24} \\ &- 0.94x_{32} - 1.42x_{54} - 2.04x_{55} + 0.96x_{74} \end{aligned} \tag{1}$$

None of the socioeconomic or demographic variables significantly influenced the perception of patients for the implementation of MR mechanism for the favourable with reservations category of the response variable. The parameter estimates of the significant independent variables from the most parsimonious model (marital status variable removed) are reported in Table 3. It can be seen from Eq. 1 and Table 3 that patients with annual income between USD 23,200 – USD 46,400 had significantly lower odds of adhering to the introduction of MR mechanism compared to those with income up to USD 11,500 ($OR = 0.24,\ 95\%\,CI:\ 0.10 - 0.56,\ p < 0.01$). Similarly, patients with annual income above USD 46,400 also had significantly lower odds ($OR = 0.13,\ 95\%\,CI:\ 0.05 - 0.31,\ p < 0.01$).

To answer R2, three variables, i.e., age, income level, and education level were found to be significant in the overall model. The statistical association between patient socio-demographic and economic attributes and the favourable category of the response variables in Eq. 2.

$$\begin{aligned} &log\left(\frac{P(Y=favourable)}{P(Y=Unfavourable)}\right)_{R2} \\ &= 2.25 - 1.19x_{24} - 0.96x_{45} - 0.81x_{55} \end{aligned} \tag{2}$$

For the favourable with reservation category of the response variable, only employment status significantly influenced willingness of patients, as shown in Eq. 3.

$$log\left(\frac{P(Y=favourable\ with\ reservation)}{P(Y=Unfavourable)}\right)_{R2} = 1.43x_{64} \tag{3}$$

The MLR based model for R3-R5 for the favourable category of the response variables are as follows:

$$log\left(\frac{P(Y=favourable)}{P(Y=Unfavourable)}\right)_{R3} = 1.80 - 1.76x_{24} - 1.94x_{25} \tag{4}$$

$$\begin{aligned} &log\left(\frac{P(Y=favourable)}{P(Y=Unfavourable)}\right)_{R4} \\ &= 1.95 - 0.67x_{12} + 0.96x_{44} + 1.15x_{45} - 1.57x_{64} \end{aligned} \tag{5}$$

$$log\left(\frac{P(Y=favourable)}{P(Y=Unfavourable)}\right)_{R5} = 2.39 - 1.02x_{24} - 1.99x_{64} \tag{6}$$

Table 3 Parameter estimates of significant variables from the MLR analysis of research questions 1–5

Referral mechanisms	Attributes	Significant variables	Significant Categories	β (SE)	z-value	Pr(>|z|)	Odds ratio (LU, UU)
MR (Baseline = unfavourable) To answer R1	F	Age	Intercept	3.11 (0.77)	4.02	0.00	22.42 (6.3, 79.57)
		Reference = 18–29 years	Age 30–39	-1.49 (0.52)	-2.85	0.00	0.22 (0.09, 0.53)
		Annual Income level	Age 40–49	-1.63 (0.53)	-3.05	0.00	0.19 (0.08, 0.46)
		Reference = Up to USD 11,500	Age 50–59	-1.36 (0.63)	-2.16	0.03	0.25 (0.09, 0.72)
		Marital status	Annual Income USD 23–46 K	-1.42 (0.52)	-2.68	0.00	0.24 (0.10, 0.56)
		Reference = Married	Annual Income >USD 46 K	-2.04 (0.53)	-3.80	0.00	0.13 (0.05, 0.31)
		Number of children	Unmarried	-0.94 (0.50)	-1.85	0.06	0.39 (0.17, 0.88)
		Reference = 0	$\geq$ 3 children	0.96 (0.58)	1.65	0.09	2.61 (1.00, 6.78)
	FWR	None of the predictors were significant					
		Δ AIC score (removed MS)	0.94%				
Noncompliance penalty (Baseline = unfavourable) To answer R2	F	Annual Income level	Intercept	2.25 (0.66)	3.38	0.00	9.48 (3.2, 28.09)
		Reference = Up to USD 11,500	Annual Income > USD 46 K	-0.81 (0.46)	-1.76	0.07	0.44 (0.20, 0.94)
		Age		-1.19 (0.45)	-2.60	0.00	0.30 (0.14, 0.63)
		Reference = 18–29 years	Age 50–59	-0.96 (0.50)	-1.91	0.05	0.38 (0.16, 0.87)
		Education level	Post-graduate degree				
		Reference = No formal education					
	FWR	Employment status	Unemployed	1.43 (0.69)	2.07	0.03	4.17 (1.34, 13.00)
		Reference = Student					
		Δ AIC score (removed MS)	0.34%				
OpR (Baseline = unfavourable) To answer R3	F	Age	Intercept	1.80 (0.78)	2.30	0.02	6.04 (1.67, 21.82)
		Reference = 18–29 years	Age 50–59	-1.76 (0.45)	-3.84	0.00	0.17 (0.08, 0.360)
			Age $\geq$ 60	-1.94 (0.56)	-3.44	0.00	0.14 (0.05, 0.361)
	FWR	None of the predictors were significant					
		Δ AIC score (removed MS)	2.53%				
Centralised EMR (Baseline = unfavourable) To answer R4	F	Education level	Intercept	1.95 (0.84)	2.31	0.02	7.02 (1.76, 27.98)
		Reference = No formal education	Undergraduate degree	0.96 (0.48)	1.99	0.04	2.61 (1.18, 5.75)
			Post-graduate degree	1.15 (0.61)	1.87	0.06	3.15 (1.15, 8.61)
		Gender	Female	-0.67 (0.39)	-1.72	0.08	0.51 (0.26, 0.97)
		Reference = Male	Unemployed	-1.57 (0.69)	-2.27	0.02	0.20 (0.06, 0.64)
		Employment status					
		Reference = Student					
	FWR	None of the predictors were significant					
		Δ AIC score (removed MS)	0.31%				
TMR mechanism (Baseline = unfavourable) To answer R5	F	Age					
		Reference = 18–29 years	Intercept	2.39 (0.75)	3.18	0.00	10.91 (3.17, 37.4)
		Employment status	Age 50–59	-1.02 (0.49)	-2.06	0.03	0.36 (0.16, 0.80)
		Reference = Student	Unemployed	-1.99 (0.62)	-3.18	0.00	0.13 (0.04, 0.37)
	FWR		None of the predictors were significant				
		Δ AIC score (removed MS)	0.12%				

F: Favourable; FWR: Favourable with reservations; β: Estimated regression coefficients; SE: Standard error; OR: Odds ratio; Lower limit (LL) and upper limit (UL) at 95% confidence interval; Δ AIC: Percentage change in Akaike Information Criteria; MS: Marital status

Higher education levels and annual income levels were associated with reluctance among participants regarding the implementation of noncompliance penalties in MR mechanisms. Patients aged 60 years or older were significantly less willing to visit other healthcare facilities in case of overcrowding at their current facility compared to patients aged 18–29 years ($OR = 0.14,\ 95\%\, CI:\ 0.05 - 0.36,\ p < 0.001$), indicating that younger patients are more likely to follow referrals to alternative facilities. Further, to determine the robustness of MLR models, the Receiver Operating Characteristic (ROC) curve with area under curve (AUC) estimates were estimated. AUC scores ranged between 74% and 84% for the fitted models.

Doctor's survey: feedback on potential new referral mechanisms

All doctors from the surveyed healthcare facilities recommended the implementation of MR mechanisms as shown in Fig. 5. None of them listed any concerns regarding its implementation across the hierarchical healthcare delivery network. Regarding the OpR mechanism,

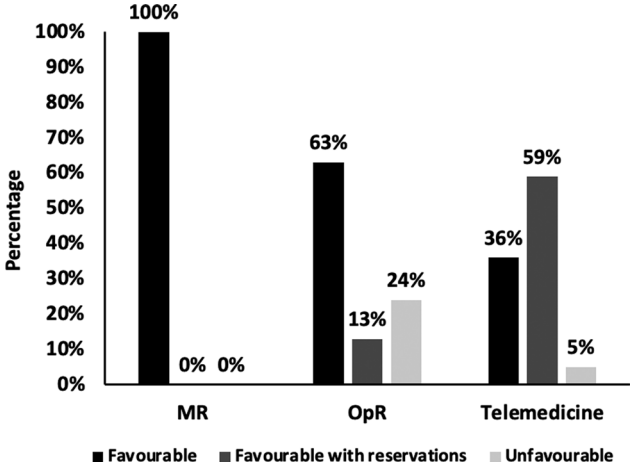


Fig. 5 Doctors' perceptions regarding potential new referral mechanisms

a significant proportion (24%) of the doctors expressed concerns about its implementation, with the major concern being the requirement of a robust technological infrastructure for information storage and sharing, lack of technical know-how, and unavailability of mobile phones with internet connectivity among patients.

Telemedicine-based consultation and referral mechanisms were favoured (36%) for the treatment of mild illnesses, for elderly patients who faced difficulty in traveling, and during infectious disease outbreaks. A majority of the healthcare providers reported concerns regarding a 'lack of personal touch' in consultations provided via telemedicine. A detailed list of doctors' concerns related to the feasibility of implementation of potential new referral mechanisms is provided in Table 4.

Discussion

This study attempts to collect and understand stakeholder perceptions of existing and potential new referral mechanisms aimed at mitigating operational challenges at public healthcare facilities, including overcrowding at tertiary hospitals and low utilization of primary and secondary care facilities. By explicitly linking findings to research questions *R1-R5*, we provide insights into how patients and doctors may respond to the introduction of potential new referral mechanisms such as such as medical referral (MR), operational referral (OpR), centralised electronic medical records (EMR), and telemedicine-based referral (TMR). designed to curb said noncompliance and mitigate operational challenges at these facilities.

Medical referral (R1)

Our survey found that a substantial proportion of patients (nearly 21%) visited higher-level healthcare facilities without an official referral from a primary or secondary healthcare facility Logistic regression analyses indicated that patients belonging to the higher income categories were more likely to bypass the MR mechanism and visit healthcare facilities of their choice ($OR = 0.13,\ 95\%\,CI:\ 0.05 - 0.31,\ p < 0.01$). This behavior indicates less trust and perception of poorer quality of services at public primary care public facilities compared to secondary and tertiary-level public facilities among economically better-off patients. Patients also directly sought care at higher-level healthcare facilities from other regions of India, as well as from other developing countries including Ethiopia and Tanzania [23–25].

Noncompliance penalties (R2)

Patients with higher education and annual income levels may have been reluctant to support noncompliance penalties in MR mechanisms because individuals with these characteristics may have fewer financial and awareness-based constraints on their health-seeking behaviour, and hence may not view favourably imposition of restrictions on the facilities they can visit for receiving care. For example, they may possess a higher degree of health literacy due to which they may wish to bypass the referral system and directly consult with specialists or seek second opinions, access hospitals with shorter wait time, and afford out-of-pocket expenses. A similar finding was reported in Bangladesh where patients residing in urban areas and belonging to middle- and high-income families displayed a higher tendency to self-refer to tertiary care facilities [26]. Conversely, patients with three or more children were more likely to favour noncompliance penalties in medical referral mechanisms as parents managing larger families are likely to have more frequent interactions with the healthcare system and a structured referral system with penalties.

Operational referral (R3)

Approximately 59% of the surveyed patients showed their willingness towards the implementation of the OpR mechanism at the healthcare network with older patients aged age 60 years or above showing less willingness towards its implementation in comparison to patients aged 18–29 years ($OR = 0.14,\ 95\%\,CI:\ 0.05 - 0.36,\ p < 0.001$). A substantial proportion of the self-directed referral patients (53%) sought care from other facilities primarily due to congestion and overcrowding which implies there is a significant scope for implementing OpR, even though many healthcare providers expressed apprehensions regarding its implementation, which included concerns regarding availability and deployment of the required technological infrastructure across facilities in the network and availability of appropriate technology such as smartphones among patients. Studies have reported the advantage of implementing OpR mechanism in minimizing average

Table 4 Concerns related to the feasibility of referral mechanisms

S.No	MR	Concerns	OpR	Concerns	Telemedicine	Concerns
1	F	Yes, this is what should ideally be there. Patients should adhere to it.	F	Personally, such data is not maintained. Requirement of technological infrastructure for the same.	FWR	This is suitable for mild illness conditions such as fever or flu. For specialized consultation, I suggest in-person check-ups.
2	F	Sometimes, there is a lack of qualified paramedical.	F	We cannot implement due to lack of data (number of patients in queue or distance between facilities).	FWR	COVID-specific cases - It should be there. General purpose – No.
3	F	No concern	F	It will save time.	FWR	Lacks personal touch, everything cannot be done virtually, physical pain cannot be felt.
4	F	Patients must adhere to it.	F	Best for everyone. Sometimes patient load is too high (very rarely but happens). Requirement of technical know-how for successful implementation.	F	No concern
5	F	Patients visit the healthcare facility of their choice which should not be allowed.	U	We do not have such system in the existing healthcare delivery system.	U	Everything cannot be treated on screen.
6	F	This is already in place.	FWR	Implementation will take 10–20 years.	F	No concern
7	F	No concern	F	It will get easier for patients and doctors.	FWR	COVID, difficulty in travelling, time limitation.
8	F	No concern	FWR	Application may stop working in between and hence robust infrastructure is required.	F	No concern
9	F	This is required.	F	No concern	F	Lacks personal touch
10	F	Very essential. Lower wait time.	F	Very good	FWR	Lacks personal touch
11	F	No concern	U	Difficult to implement (lack of technical know-how among patients).	FWR	Lacks personal touch
12	F	No concern	F	No concern	FWR	No concern
13	F	No concern	F	No concern	F	An alternative for healthcare delivery.
14	F	No concern	U	Implementation will require time and effort	FWR	No concern
15	F	No concern	UF	Global citizen. Patient chooses to visit a given healthcare facility. I cannot ask to go and visit other facility.	FWR	Beneficial: remote areas, far-flung areas: boon, metropolitan: slightly far-fetched compliance; Disadvantages: everyone has become a doctor.
16	F	This system is in the existing delivery system.	U	There are already multiple facilities, so the load gets distributed. Wait time does not exceed 15 min	FWR	Mild condition, follow up, chronic illness, bedridden – favourable
17	F	No concern	FWR	Concerns about accuracy of estimated wait time or length of stay estimates	FWR	Mild diseases (Not for severe); examination required (severe cases)
18	F	No concern	F	Balances patient load. Sometimes we are overburdened	F	No concern
19	F	Similar to the UK and the Canadian health delivery system, India should have something like this	F	PAN India hospital network should be also there	F	Telemedicine can be used for follow-up and not for the first consultation.
20	F	Lack of qualified paramedical and lesser number of medical staff.	F	No concern	FWR	COVID specific cases. Not for general day-to-day consultations
21	F	Proper management of medicine. Patients ask for referrals to other facilities	F	No concern	F	For mild conditions
22	F	No concern	F	No concern	FWR	Strong internet connectivity

F: Favourable; FWR: Favourable with reservations; UF: Unfavourable

time to receive care across healthcare facility networks [12, 27]. While smartphone penetration in India – especially in urban areas – is extensive, a formal investigation of its access among patients visiting such facilities may be needed [28].

Centralised EMR (R4)

Nearly ~47% of the referred patients had to undergo repeat laboratory investigations even if a recent diagnostic report was available. This could be attributed to a lack of trust among care providers at higher-tier facilities for the procedures carried out in other facilities, including lower-tier facilities. If quality of care is not a concern at lower-tier facilities, then this finding points to the need for appropriate sensitization of care providers at higher-tier facilities. Since nearly all the surveyed doctors and over 75% of patients favoured the implementation of a centralised EMR, healthcare administrators may consider investing in technological infrastructure that can store, update, and share patients records across different levels of care in India, while also addressing concerns regarding data privacy. This recommendation also aligns with the MLR findings from *R4*, which showed that female patients were less likely to favour centralised EMR ($OR = 0.51,\ 95\%\ CI:\ 0.26 - 0.9,\ p = 0.08$).

Telemedicine based referral (R5)

Telemedicine based consultation and referral could bridge gap between rural and urban healthcare delivery systems and may serve as an alternate consultation tool for patient care, as emphasized by the surveyed doctors. While many patients expressed apprehensions; MLR results indicated that elderly patients were less likely to favour this mechanism ($OR = 0.36,\ 95\%\ CI:\ 0.16 - 0.80,\ p < 0.05$) in comparison to younger patients. Nonetheless several doctors emphasized that telemedicine could particularly benefit older patients who face challenges in travelling to distant facilities. Effective implementation, however, depends on a robust information and communication technology infrastructure, with its adoption influenced by factors such as internet connectivity and the technical proficiency of both patients and care providers [29].

Additional finding on healthcare seeking behaviour

A majority of patients (nearly 83%) who chose self-referral visited private healthcare facilities owing to the timely provision of care and adequate time spent by healthcare providers with the patient despite the significantly higher overall treatment costs. Patients are either compelled to, or prefer to, seek care from private providers, often at a cost beyond their paying capacity [15]. Kiwuwa et al. [30] found similar behavior when exploring causes for delays in the diagnosis of tuberculosis treatment at a referral center in Uganda, where a significant proportion of patients (39%) visited private clinics more commonly than government health facilities owing to lesser delays. Private practitioners can play a crucial role in bridging the healthcare gap in India's mix healthcare system, improving general access and service quality, reducing wait time, and relieving overburdened public facilities. Achieving appropriate balance and coordination between the public and private sectors is likely to be essential for achieving Universal Health Coverage (UHC) [31, 32].

Recommendations for the Indian public healthcare administrators

To strengthen the referral pathways and improve care coordination across the hierarchical healthcare delivery system, the following recommendations are proposed for Indian public healthcare providers.

- **MR with triage and penalties** Healthcare policymakers can consider starting stakeholder engagement and sensitization campaigns on the clinical, operational, and financial consequences of bypassing lower-level facilities for healthcare. Delays in receiving medical care due to a shortage of medical resources also exacerbate noncompliance with referral mechanisms; hence policy makers can look for strategies such as increasing resource capacity – for example, via the early discharge of medically stable patients.

- **EMR rollout with privacy safeguards** Deployment of a centralised EMR system can aid doctors in diagnosing and treating patients in a better manner by making patients' medical histories more easily accessible to the doctors. The Indian government's recent initiative of providing every Indian national a unique Health ID called ABHA (Ayushman Bharat Health Account), an electronic repository of all health-related records is a welcome step towards this. Effective use of the ABHA ID at all levels can be encouraged, which may include sensitization and adequate training for all medical personnel involved.

- **OpR readiness investments** Policymakers can consider the implementation of OpR to reduce congestion and ensure equitable utilization of medical resources at public facilities. The NHSQuicker web/mobile application provides up-to-date wait time information for the five closest emergency departments in the South-West region of England, helping patients make more informed decisions regarding which healthcare facility to visit [33]. Similar digital innovations, alongside adequate training for doctors and patients, may improve health outcomes in India. Recent reforms under the Ayushman Bharat Program (ABP) - upgrading 150,000 Health and Wellness Centres (HWCs) and expanding access to hospitalization

Table 5 Synthesis table

Potential mechanism	Key findings	Policy implications
Medical Referral	a substantial proportion of patients visited higher-level healthcare facilities without an official referral; patients belonging to the higher income categories were more likely to bypass ($OR = 0.13,\ 95\%\ CI:\ 0.05 - 0.31,\ p < 0.01$).	Introduce triage; penalties, strengthen referral pathways
Electronic Medical Record	repeat tests were common; female patients less likely to favour centralised EMR ($OR = 0.51,\ 95\%\ CI:\ 0.26 - 0.9,\ p = 0.08$).	Rollout EMR with privacy safeguards, train staff
Operational Referral	59% of the surveyed patients showed their willingness; Many healthcare providers also expressed apprehensions regarding its implementation,	Pilot digital wait time tools; address provider concerns
Telemedicine based referral	elderly patients were less likely to favour this mechanism ($OR = 0.36,\ 95\%\ CI:\ 0.16 - 0.80,\ p < 0.05$)in comparison to younger patients	Invest in digital literacy; ICT infrastructure

services at secondary and tertiary care for bottom 40% of total population through PM-JAY are important steps towards strengthening primary care and reducing overloading at higher-level facilities [2, 15]. We present a synthesis of the referral mechanisms, key findings from the stakeholders, and policy implications in Table 5.

A key limitation of this study is that the health-seeking behavior analyses are based solely on surveys conducted in the South-west Delhi district. *Consequently, the findings are likely to find resonance among most urban and in particular metropolitan Indian regions.* Additionally, only patients who received care at the surveyed public facilities were interviewed; patients attending private healthcare facilities, tertiary hospitals, or households were not included. The study findings are based on a survey conducted in a single (albeit densely populated) district of a major Indian city. Given the vast and diverse healthcare delivery landscape in India, similar studies may be conducted at both private and public healthcare facilities across the country to generate quantitative evidence applicable to the Indian population at large. The survey questionnaires can be expanded to include multiple stakeholders within the healthcare ecosystem, including patients, healthcare providers, administrative staff, and policymakers. Even though the facilities we surveyed were all in south-west Delhi, our findings are likely to be at minimum applicable to urban Indian cities because: (a) the health systems are similar in most Indian cities [34]; and (b) bypassing is a pan-Indian phenomenon, etc [35].

Conclusion

Our study provides an example for how stakeholder perceptions may be collected and analyzed prior to the (modification) deployment of (existing) new referral mechanisms. It also illustrates the benefits of conducting such an analysis in terms of the insights generated into the associations between patient behavior, their demographics and perceptions of different referral mechanisms, which may in turn influence their compliance. with said mechanisms. If the proposed referral mechanisms are thoughtfully implemented and supported by adequately resourced facilities, there could be potential improvements. Future research directions could include potential computer simulation modelling of the proposed referral interventions to analyse their outcomes on the performance of the healthcare facility network.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12913-025-13602-y.

Supplementary Material 1

Acknowledgements
We thank the Directorate General of Health Services (DGHS) for permitting us to conduct surveys across the public healthcare facilities in the Southwest Delhi district in India.

Author contributions
NF and VR designed the study. NF administered the survey and collected the stakeholder responses. NF, KK, and VR developed the study objectives. NF conducted the statistical analyses. NF wrote the manuscript with support from VR and KK. VR supervised the project.

Funding
This work was supported by the Prime Minister's Research Fellowship (PMRF), Government of India with the grant number PL/N08.

Data availability
The data that support the findings of this study are available from the corresponding author, NF, upon reasonable request.

Declarations

Ethics approval and consent to participate
The ethics committee of the authors' institute approved the study protocols with approval number IITD-IEC-ID-P064. Participants provided their consent to participate by filling out participant informed consent forms. The study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki.

Consent for publication
NA.

Competing interests
The authors declare no competing interests.

Received: 22 April 2025 / Accepted: 6 October 2025
Published online: 30 October 2025

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